Human-centered design approach to support IT professionals: A systematic literature review

Raissa Desyandita Tedjomurti
Department of Electrical and Information Engineering
Universitas Gadjah Mada
Yogyakarta, Indonesia
raissadesyandita@gmail.com

***Abstract*— In recent years, Human-Centered Design (HCD) has drawn growing interest in software engineering as a practical way to create systems that genuinely fit the people who use them. Yet, discussions on how HCD actually supports software developers and IT professionals are still scattered and lack synthesis. This study conducts a Systematic Literature Review (SLR) of thirty studies released between 2020 and 2025 to understand how HCD principles have been used in software development practices. Most of the reviewed studies apply qualitative and design-oriented methods that rely on close user involvement, iterative prototyping, and feedback-driven improvement. Five main frameworks appear frequently as reference models. Across these studies, HCD is shown to enhance usability, inclusivity, and communication among teams, although practical barriers such as time pressure, limited expertise, and the fast pace of agile projects still make implementation uneven. Altogether, the review suggests that HCD should be viewed not only as a design method but as a collaborative mindset that helps teams build technology more responsibly and with greater awareness of human values.**

Keywords—human-centered design, software engineering, systematic literature review

I. INTRODUCTION

Human-Centered Design (HCD) comes from a growing acknowledgement that technologies and systems should be customized to meet human needs, limitations, and behaviors, rather than forcing users to adapt to technological limitations. This perspective originated in the mid-20th century from the domains of ergonomics, cognitive science, and software engineering approach [1]. Researchers began to study how people interacted with complex systems, like industrial machines and early computer interfaces, to make them safer, easier to operate, and more enjoyable overall. This change in focus meant that the goal was no longer just to optimize the technology, but also to make sure that users were comfortable, understood, and pleased. As digital technology became more and more significant, it was important to investigate how people perceive, think, and feel so that systems could be made that were easy to use and understand.

HCD slowly became a systematic design paradigm, profoundly influenced by Norman [2], who made the idea of user-centered design prominent in the late 1980s. This method encourages users to empathize with each other, evaluate things over and over, and be involved in the process of making or developing something. HCD is different from other design methodologies because it doesn't focus on technical or aesthetic difficulties. Instead, it focuses on understanding the real-world problems and situations that people face. This "human first" approach of thinking has become very important in many fields, including product design, software engineering, healthcare, and social impact projects. It makes sure that technology is based on what people think and feel. HCD always leads to the creation of solutions that improve the user experience and make it easier for people to embrace new systems by focusing on how technology design fits with human values and experience.

Within the realm of Human-Computer Interaction (HCI), IT professionals confront several problems to meet the demands and requirements of users. The integration of human-centered design into the software engineering process has been contemplated for more than ten years [3]. HCD seeks to generate precise solutions to diverse design challenges. Their inclination is to address user requirements through their design solutions. Nevertheless, some firms that have engaged in HCD within their software development processes continue to encounter challenges stemming from inadequate interface and usability of their products [3]. This also affects the mental health of IT professionals. The Software Engineering (SE) domain is not exempt from the impact of workplace mental health, as evidenced by recent studies of IT professionals.

This study aims to thoroughly assess and synthesize human-centered research, providing a comprehensive overview of the area, including the identification of prevailing trends and prospective directions for HCD within the SE domain, particularly for IT professionals. The present paper is divided into the following parts: Section 2 presents the main results of earlier Systematic Literature Reviews (SLRs) on similar topics. Section 3 explains the method that was used. Section 4 shows what the author found out about the research questions that the author set up in the methodology. Section 5 outlines the conclusion, summarizing the main findings about the current problems and possible ways to improve the sector.

II. RELATED WORK

This section presents a relevant Systematic Literature Review (SLR) or Systematic Mapping Study (SMS) concerning HCD subjects. To the known knowledge, no extensive study concentrates on IT professionals. Consequently, we investigated appropriate sources, including systematic literature reviews, that remain relevant to the software engineering domain. The author also highlights the SLR pertaining to the human-centered approach within non-software engineering domains.

A. Secondary studies of HCD in SE domain

A number of secondary studies have examined how HCD principles are integrated into the software engineering process. Ali [4] examined Human-Centric Software Engineering (HCSE), highlighting the significance of empathy and user orientation in the software development process. Sharma et al. [5] analyzed various Human-Centered Software Development (HCSD) methodologies, demonstrating that Design Thinking commonly aligns with Agile methods but frequently lacks standardized evaluation protocols.

Adhikari [6] proposed a set of guidelines for embedding HCD practices into software lifecycles, highlighting the imperative of collaboration between developers and designers to enhance usability and productivity. In related discussions, Auernhammer [7] explored Human-Centered AI (HCAI) as a framework that merges ethical responsibility with SE practices, while Constantinides et al. [8] analyzed how new regulatory expectations push developers toward more transparent and accountable AI-driven systems. Nonetheless, they also point out recurring obstacles such as insufficient empirical validation, limited organizational support, and the ongoing difficulty of embedding HCD into Agile or DevOps environments [5][6]. Overall, the reviewed studies in the SE domain highlight that HCD has the potential to change how software is developed, though they also indicate that it has not been fully explored how it affects the work of IT professionals.

B. Secondary studies of HCD in other SE domain

HCD has been thoroughly explored in the healthcare, education, and other industrial sectors, in addition to software engineering. Göttgens [9] studied the application of HCD and Design Thinking in healthcare, revealing that evaluation methodologies were often inconsistent and poorly documented. Vial [10] conducted a thorough evaluation of digital mental health therapies, noting that interactive methods were commonly utilized, yet not subjected to rigorous assessment. Zhai et al. [11] expanded upon earlier frameworks that combined HCD with implementation science to connect the design phase with long-term adoption outcomes. Stiles-Shields [12] and Nijagal [13] broadened HCD to include equity-centered healthcare initiatives, illustrating how participatory design can uncover linguistic and cultural barriers in patient care.

Molina [14] and Levander [15] demonstrated that co-design enhances teamwork, accessibility, and trust within clinical settings. Ngoc [16] examined the significance of HCD in Industry 4.0 systems, highlighting human-machine collaboration, whereas Sajid [17] assessed human-centered robotics design, concentrating on ergonomics, user trust, and safe interaction. The evaluations demonstrate that participatory HCD methods consistently improve usability and engagement, albeit primarily focusing on particular applications with limited longitudinal validation [9][10].

This study aims to fill the existing gap by examining the implementation of Human-Centered Design (HCD) principles in software engineering to support IT professionals, merging theoretical frameworks with tangible and measurable outcomes.

III. Methodology

In this section, the author followed the general principles set forth by Tulili [26], which were derived from the instructions of Petersen et al. [18]. The author delineates the research questions or objectives in Section A, the identification of the recovered studies in Section B, the quality assessment of the obtained studies in Section C, and the data extraction process in Section D. Finally, the data synthesis is shown in Section E.

The author established a comprehensive review process, executed thorough searches, filtered the studies, and performed data extraction. After that, the author conducted data analysis and examined the data to present it in tables, graphs, and a preliminary analysis report.

A. Research Questions (RQ)

To aid researchers in understanding the use of Human-Centered Design (HCD) and to improve the software development process and support IT professionals, we offer five research questions:

- **RQ1**: What categories of research have been undertaken to investigate the application of HCD in software engineering and information technology environments?
  Rationale. This inquiry seeks to deliver a comprehensive review of the current research environment regarding HCD applications in software engineering. Identifying study types facilitates the recognition of research trends, methodological inclinations, and potential deficiencies that inform future directions in this field.
- **RQ2**: What types of data sources have been used to analyze the integration of HCD principles into the software engineering process?
  Rationale. This inquiry seeks to examine the variety of data sources utilized by researchers (i.e., case studies, user feedback, or experimental data). It is used to clarify how empirical evidence supports the implementation of HCD in improving software development and assisting IT professionals.
- **RQ3**: What methods and frameworks have been applied to implement or evaluate HCD practices in the context of software engineering?
  Rationale. This inquiry seeks to ensure the methodologies, concepts, and frameworks utilized to implement HCD within software engineering. By delineating these methodologies, we can comprehend how various study strategies facilitate the practical implementation and evaluation of HCD in aiding IT professionals.
- **RQ4**: What are the observed benefits and challenges of using HCD in software engineering projects involving IT professionals?
  Rationale. This inquiry seeks to understand the positive outcomes and limitations of HCD adoption for evaluating its real-world impact. It helps identify how HCD influences aspects (i.e., usability, productivity, collaboration, or satisfaction) among IT professionals.

While it also reveals barriers that may hinder its integration into SE practices.

- **RQ5**: How does the integration of HCD influence organizational practices and team processes in software engineering environments?
  Rationale. This inquiry seeks to explore the broader effects of adopting HCD beyond individual usability outcomes. By examining its influence on collaboration, communication, decision-making, and workflow management among IT teams, it helps reveal how HCD contributes to shaping a more adaptive and user-oriented organizational culture within software development settings for adoption for evaluating its real-world impact.

B. Relevant Study Identification

To identify the relevant studies for this SLR, the author defined the search query from the research questions formula. Thereby, the search query is: "human-centered *design*" AND ("*software engineering*" OR "*software development*").

The author used several reliable online database collections, such as IEEE Xplore Digital, Springer, Sage, ResearchGate, ACM Digital Library, and Elsevier. However, it is not possible to use other databases like JMIR, Taylor & Francis, and Wiley Online Library. The author utilized the search functionalities in each online library, employing the keywords of "Full Text" or "Download this PDF". The aggregation of results from all these libraries yielded 369 papers.

Furthermore, the collection of papers was filtered manually, based on the inclusion and exclusion criteria summarized in Table 1. The author derived the inclusion and exclusion criteria from the research questions, and it generated 4 acceptable paper criteria and 2 not acceptable paper criteria. All of the selected papers have been selected from the databases mentioned above to be identified in the filtering phase.

TABLE I. INCLUSION AND EXCLUSION CRITERIA

Inclusion Criteria	
IC1	The paper must be a peer-reviewed paper and published at conferences, workshops, or journals.
IC2	The paper must be accessible online (i.e., PDF available in the selected database collection or through Google).
IC3	The study must be published between the years of 2020 and 2025.
IC4	The study reported must be related to a human-centered approach.
Exclusion Criteria	
EC1	The non-English paper includes the non-English data source used.
EC2	The paper is not a full research publication (e.g., abstract-only submissions, presentations, tutorials, posters, or surveys).

The author applied the inclusion and exclusion criteria during the filtering phase, resulting in 49 papers identified for consideration in the initial set and as the input for the selection phase. In the selection phase, if the paper's title explicitly used the keyword "*human-centered*", the author looked more in detail at the abstract. If the abstract's information indicated a study about implementation with a human-centered approach, the paper was passed to the next phase. If the keyword was not found in the title, the author still read the abstract, skimming by looking for information such as the purpose, the method, the results, and the conclusion that would indicate the focus of the paper on a human-centered approach.

After completing the filtering phase, 21 papers were obtained. These papers (Start Set) are added into the snowballing process, as proposed in the guidelines [18]. Backward Snowballing is to examine the list of references of the papers gathered in the previous step. If the reference was already in the Start Set, it was ignored. If not, the author looked at the title, abstract, and relevant parts of the candidate paper to decide if it should be included in the next phase.

Meanwhile, Forward Snowballing means to examine the candidate papers based on the citations to each paper in the Start Set. The author used Elsevier to find scientific literature that cites the base papers. Subsequently, it looked at each manuscript that cited the Start Set papers to see if the candidate paper was part of it. If the reference was already in the Start Set, it was ignored. If not, a similar process was used during the filtering phase.

After using the snowballing phase, the author found 30 new papers (Final Set). In conclusion, the author looked at 310 that were published between 2020 and 2025, with 21 of them being published in the first half phase and 9 in the second half phase.

C. Study Quality Assessment

The author identifies the suitable papers to answer the research questions described above with the quality assessment by Kitchenham [19]. The assessment criteria form is presented in Table 2, containing a list of questions to assess each of the studies.

TABLE II. ASSESSMENT CRITERIA FORM

No	Questions	Focus
1	Are the research objectives strongly defined about human-centered design?	RQ1
2	Does the research clearly state the methods applied (e.g., qualitative, case study, experimental, or mixed)?	RQ3
3	Does the research explicitly implement or weigh human-centered design principles, frameworks, or methodologies (e.g., user research, prototyping, co-design)?	RQ1, RQ3
4	Does the research evaluate at how usability, satisfaction, or productivity of IT professionals affects the results?	RQ4
5	Is it clear how the data was collected?	RQ2
6	Are the data sources (e.g., case studies, surveys, project artifacts) clearly identified and justified?	RQ2
7	Does the study report benefits, limitations, or challenges in implementing human-centered design in software engineering?	RQ4, RQ5

D. Data Extraction

Upon comprehensive evaluation of the complete text of each primary study, the author extracted the data from the paper. The author gathered qualitative and quantitative data

from the studies: the qualitative data included of variables, methods, type of research, data sources, procedures, results, and conclusions. The quantitative data included the number of participants or the sample size, the number of data sources, and the dimensions of datasets.

E. Data Synthesis

After completing the data extraction stage, the author carried out a thematic and descriptive synthesis to combine and interpret the results from all selected studies. The studies were organized based on their main focus, research methods, and the HCD frameworks applied. Through repeated comparison, the author identified recurring ideas such as usability improvements, stronger user engagement, better teamwork, and common challenges in implementing HCD. This synthesis process helped provide a clearer picture of how Human-Centered Design has been applied to enhance software engineering practices and support IT professionals.

The synthesis results were arranged based on the five research questions of this study. To begin with, for the types of studies (RQ1), most papers used qualitative or design-oriented approaches such as case studies, participatory design projects, or framework discussions. Only a few applied mixed methods or systematic reviews, which shows that HCD research often focuses more on understanding user experiences than on testing strict hypotheses.

Regarding data sources (RQ2), the reviewed studies mainly collected information from interviews, co-design workshops, questionnaires or surveys, and usability testing sessions. These allowed researchers to gain deeper insights into user interactions with systems and their requirements. It also emphasizes the importance of involving users directly throughout the design process.

When examining the methods and frameworks (RQ3), the predominant approaches identified were participatory design, co-design, design thinking, and iterative prototyping. Some papers also tried to combine HCD with agile development or implementation frameworks to make it easier to connect user feedback with the software process.

For the benefits and challenges (RQ4), the studies agreed that HCD leads to better usability, stronger user satisfaction, and smoother teamwork between designers and developers. However, they also pointed out issues like limited time, lack of resources, and the difficulty of fitting HCD methods into fast-moving engineering workflows.

Lastly, in exploring the broader impact of HCD (RQ5), the findings suggest that HCD can encourage empathy, better communication, and shared responsibility within project teams. Even so, consistent adoption remains tricky, especially in organizations where team members are still unfamiliar with HCD principles or lack managerial support. Overall, these findings show that Human-Centered Design helps bridge the gap between people and technology. It makes the software development process more flexible, open to everyone, and in line with what users—including IT professionals—really need to do their jobs.

IV. Results and Analysis

A. Overview

In this section, we gathered data synthesis of selected studies that explained the use of Human-Centered Design (HCD) in software engineering and IT-related contexts. The focus is on how HCD has been used to improve user experience, usability, and collaboration throughout the system development process. The findings are organized based on the five research questions (RQ1-RQ5), followed by a short discussion of what these results mean in a broader context.

B. Findings

[RQ1—What types of studies have explored HCD within SE and IT work context?]

Most of the reviewed studies used qualitative or design-oriented approaches, such as case studies, participatory design research, and conceptual framework development. A smaller portion combined different methods—mixing surveys, observations, and usability evaluations—while only a few were large-scale reviews. Overall, the trend shows that research on HCD in software and IT tends to focus more on exploring real-world applications and user experiences rather than statistical or experimental testing.

[RQ2—What kind of data sources have been used to study HCD implementation?]

The data sources found across the studies were mostly interviews, co-design workshops, surveys, focus groups, and usability testing sessions. These methods helped researchers collect direct input from users, designers, and developers to better understand how systems are used and what improvements are needed. A few studies, especially in AI and virtual reality projects, also analyzed observation notes and system data. This pattern shows that HCD research strongly values first-hand feedback and close involvement with end users.

[RQ3—What methods and frameworks have been applied to implement or evaluate HCD practices?]

In terms of frameworks, the most frequently used were Participatory Design, Co-Design, Design Thinking, and the IDEO model. Numerous studies integrated these methodologies with Agile, Design Science Research, or Implementation Science to enhance the adaptability and collaboration within the design process. Some studies have looked to make AI more ethical and easier to use. These findings illustrate the versatility of HCD across various domains and technical contexts. Here is the distribution of HCD frameworks used for all papers in Table 3 and Figure 1.

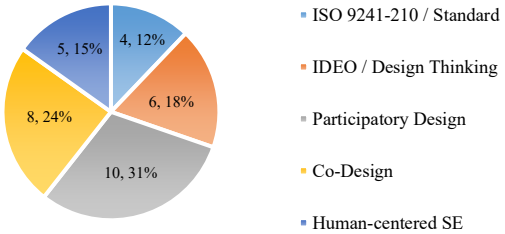


Figure 1. Number of Each HCD Framework Papers

TABLE III. DISTRIBUTION OF HCD FRAMEWORKS

HCD Framework	Relevant Paper	Main Purpose
ISO 9241-210 / Standard	[4], [6], [11], [20]	Provides a structured process for understanding users, designing, and evaluating usability
IDEO / Design Thinking	[5], [9], [15]	Promotes empathy, ideation, and iterative prototyping to generate user-centered solutions
Participatory Design	[10], [12], [16], [21]	Encourages collaboration between users and designers to ensure inclusivity and contextual fit
Co-Design	[10], [12], [14], [22]	Focuses on shared creation and decision-making between stakeholders during design
Human-Centered of Software Engineering / Artificial Intelligence	[4], [7], [23], [24], [25]	To embed HCD principles into SE workflows and system life cycles

[RQ4—What are the main benefits and challenges of adopting HCD in SE and IT works?]

Almost all studies reported positive outcomes from adopting HCD, such as better usability, higher user satisfaction, improved inclusivity, and stronger collaboration between users and developers. HCD also helped teams better understand user needs and create designs that match real work contexts. However, several obstacles were mentioned, including tight project timelines, limited budgets, and difficulties fitting HCD into fast-paced development cycles. Another recurring issue was that many technical teams lacked training or experience in using HCD systematically.

[RQ5—How does the integration of HCD influence organizational practices and team processes?]

From an organizational standpoint, HCD was shown to enhance communication, collaboration, and empathy among team members from diverse backgrounds. It encouraged shared ownership and working together to solve problems during the whole development process. Companies had different ways of using HCD. Some made it a part of their daily work, while others saw it as an extra step. These disparities were frequently associated with leadership support, the company's culture, and the extent to which teams comprehended the long-term advantages of HCD.

C. Discussion

The overall findings show that HCD is increasingly seen as an important part of how modern software and systems are developed. By focusing on user needs from the very beginning, HCD helps reduce the gap between technical solutions and real human experiences. When users are actively involved throughout the process—through interviews, workshops, or usability testing—the final product tends to be easier to use, more inclusive, and better suited to real-world work situations.

Nonetheless, the assessment also revealed other problems. Numerous studies indicate that implementing HCD in reality can be challenging, particularly in rapid software development cycles characterized by constrained time and resources. While most studies revealed short-term enhancements, including improved usability and increased user engagement, only a limited number assessed the long-term effects of HCD on productivity, team dynamics, or software adoption. The enduring impact of HCD on organizational performance remains inadequately comprehended.

Another key issue is that HCD should be seen not merely as a design methodology but as a cognitive framework that influences team collaboration and decision-making processes. When HCD is used early in the software development process, it helps people understand each other better, communicate better, and share responsibility for results. Still, companies do not always do things the same way. Teams that do not know much about HCD often have trouble using its methods effectively, but those that have the support of their managers and enough training are more likely to do well.

V. CONCLUSION

The review of thirty studies shows that applying Human-Centered Design (HCD) plays a crucial role in developing software that genuinely reflects user needs and expectations. The synthesis revealed five dominant frameworks—Participatory Design, Co-Design, Design Thinking, ISO 9241-210, and HCSE / HCAI—each emphasizing empathy, iteration, and collaborative creation. Overall, the studies indicate that using HCD leads to better usability, greater user satisfaction, and more inclusive outcomes. Yet, persistent obstacles remain, including limited time, insufficient expertise, and the ongoing challenge of aligning HCD principles with agile development environments.

At its core, HCD is more than just a design method. It is a way of thinking that promotes understanding and working together to solve problems in technical work. It helps make systems that are not only efficient and easy to use but also morally aware and responsive to the needs of society when used from the start of development.

VI. FUTURE WORK

Future research ought to focus on providing more comprehensive data regarding the influence of HCD on software quality and ongoing user engagement. Most current research relies on qualitative methodologies. Thus, incorporating quantitative assessments—such as usability testing, adoption rates, or user performance metrics—could more definitely demonstrate the efficacy of HCD. Expanding this research into nascent technological fields, including artificial intelligence, automation, and data-driven systems, would enhance the protection of human values in rapidly evolving developmental environments.

Future research also ought to investigate pragmatic approaches for integrating HCD principles into routine software development processes. Organizations could use HCD more consistently and on a larger scale. It can be made with clear and easy-to-use frameworks like training programs or design toolkits. By integrating ethical sensitivity with technological advancement, forthcoming studies can bolster

the status of HCD not merely as a design philosophy but also as a significant method for assessing software efficacy and user impact.

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